

📅 2026 LLM Architect Roadmap Phase 1: The Mathematical Foundations (Months 1–2) Focus: Linear Algebra, Calculus, and Probability. Before you code a transformer, you must understand how Weights and Biases interact.

Primary Course: Mathematics for Machine Learning and Data Science (DeepLearning.AI)

What you learn: Tensors, Eigenvalues (for embeddings), and the Chain Rule (for backpropagation).

Cost: ~\$49/mo (Financial aid available).

Milestone: You can manually calculate the gradient of a simple 3-layer network on paper.

Phase 2: Neural Networks "Zero to Hero" (Months 3–4) Focus: Building the "Brain" from scratch in Python/PyTorch.

Primary Resource: Andrej Karpathy's Zero to Hero (Free / YouTube).

What you learn: You will build Micrograd (a tiny autograd engine) and Makemore (a character-level language model).

Cost: Free.

Milestone: A locally running GPT-style model that you wrote line-by-line without using import transformers.

Phase 3: Transformer Architecture & NLP (Months 5–6) Focus: The "Attention is All You Need" mechanism.

Primary Course: Hugging Face NLP Course (Industry Standard).

Secondary Course: Natural Language Processing Specialization (DeepLearning.AI).

What you learn: Multi-head attention, Positional encoding, and Tokenization strategies.

Cost: Free (Hugging Face) / ~\$49/mo (Coursera).

Milestone: Fine-tune a Llama-3 or Mistral model on your own dataset using LoRA/QLoRA.

Phase 4: Building & Pre-training Your Own LLM (Months 7–9) Focus: Dataset curation and custom pre-training (SLMs).

Primary Resource: DeepLearning.AI: Generative AI with LLMs.

Applied Project: Create a "Small Language Model" (SLM) trained specifically on art history and wall-decor trends for your GYWA project.

Cost: ~\$49/mo + ~\$50 in GPU compute (Google Colab / RunPod).

Milestone: You have a custom-built model that out-performs GPT-4 on specific art-related queries.

Phase 5: Professional Certification & MLOps (Months 10–12) Focus: Validation and Production-grade deployment.

Certification Choice 1 (Hardware): NVIDIA Certified Associate – Generative AI (~\$125).

Certification Choice 2 (Cloud): Microsoft Exam AI-300 (This is the 2026 successor to DP-100/AI-102 for MLOps).

What you learn: Deploying LLMs at scale, RAG pipelines, and Vector Databases.

Milestone: You are a Certified AI Engineer.